

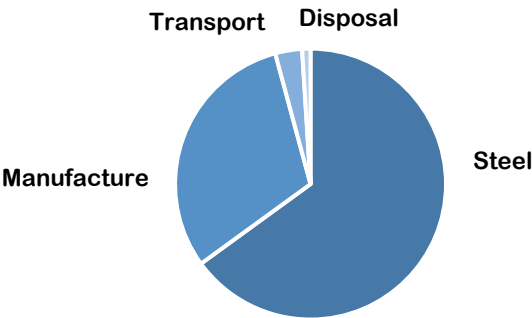
ModuSHAVE



The problem

Environmental problems

An estimated 2 billion razors are disposed of in the US alone every year.



- Polypropylene takes about 20-30 years to decompose.
- An estimated 1% of polypropylene is recycled.
- Estimates suggest that PP from recycling emits 88 per cent less CO₂ than producing it from scratch
- Polypropylene recycling is expensive and difficult to carry out.

0.0957 kg CO₂ emitted per razor



736m driven

Assuming 52 razors per year



31.8km driven

10000 kg CO₂ per person per year



77km driven

Even though polypropylene and carbide steel are recyclable alone, they are unable to be recycled when together. Most razors are very difficult to disassemble, especially around the cartridge due to safety reasons, meaning the materials cannot be separated and therefore not recycled.

Systematic issues



Unclear instructions

Few razor brands have easy to understand instructions about how to recycle their product. This means many users either recycle their razors at home, or just don't attempt to recycle them. Both of these options result in the razor not being recycled.



Ease of recycling

One way to recycle razors would be through TerraCycle: where you send off your razors in an envelope to be recycled. Unfortunately, there is a long wait to receive your prepaid envelope and this may deter users from recycling. Recycling razors should be more convenient.

Societal problems

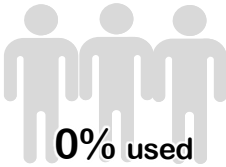
Among users interviewed:



75% prefer to use disposable razors



83% don't consider sustainability when shopping



0% used TerraCycle's recycling scheme

Multiple users admitted to throwing away razors before the end of their lifespan, saying:

"I don't really know when I can't use it anymore so I just throw it out."
- Emilia

"I go through quite a lot of them (razors) because I want a close shave and also I feel like I'll cut myself if it's used up. I don't really know when it's actually used fully though."
- Gladys

If human error was reduced, this would reduce the overconsumption of razors meaning less could be manufactured, less will end up in landfills and there will be less stress on the recycling system.

Also, none of the users had heard of the TerraCycle recycling scheme, which suggests there isn't enough education about the proper recycling of razors. Further evidence for this is that a reason why many of these users don't consider sustainability when shopping is because they believe their non-sustainable razors are recyclable.

How might we?

- ? Make disposal instructions clearer on packaging?
- ? Increase the number of recyclable elements in the system?
- ? Increase user incentive to recycle?
- ? Increase re-use but maintain safety?
- ? Reduce the carbon footprint of the packaging?
- ? Integrate ease of disposal with sustainability?
- ? Keep the product at a competitive price?
- ? Reduce the energy loss caused by transportation?

Opportunities

- ! Instructions of the disposal process on the box itself. QR code link to the website with thorough instructions.
- ! The razor body is durable and long-lasting so the user can keep it for life.
- ! The packaging is made of a sustainable material.
- ! The packaging can be used to store obsolete razors before recycling.
- ! The razor can be disposed of easily as the user does their weekly activities.
- ! A discount as incentive for the user to recycle.
- ! A product service system that sends users the right amount of razors for their needs and schedules to reduce overconsumption.
- ! An app alongside the PSS so users can reschedule, find information, and find user ID for their discount.

Future Scenario and Time Horizon

2022

The scenario for our product service system is set 10 years in the future. In this time it is assumed that both consumers' and the government's attitudes towards sustainability changes, along with various technological advancements.

These changes are outlined below:

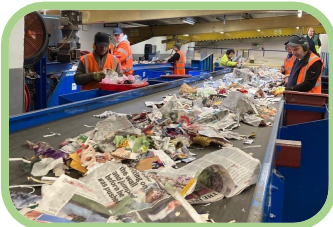
Decreased Delivery Emissions

Through the increased development of alternative, more renewably fuelled vehicles, the amount of CO2 emissions per delivery is greatly reduced. Also, due to advancements in technology there is greater optimisation for delivery routes.



Materials and Recycling

Recovery centres become widely available in all countries. Government legislation is introduced restricting the use of plastics and excessive packaging. This leads to the development of more smart packaging.



Environmental Responsibility

In the future, consumers become increasingly more eco-conscious. With greater education and awareness about the climate crisis and resources running out, sustainability is a key focus for consumers. Waste becomes much less socially acceptable.



Internet of Things

There is a development in the syncing and processing of data. Most devices become smart and can easily be synced. This means users are seamlessly provided with information through different interfaces and can become more involved in unseen processes like recycling by scanning an ID and receiving updates.



Cost of sustainability

Technological advancements in recycling and government funding into sustainable practices means that sustainable products are regularly affordable.



A future persona was also created to show how ModuSHAVE will fit into the future society and will improve the consumer experience whilst maintaining sustainability.



Future Persona



Name: Linda

Age: 20

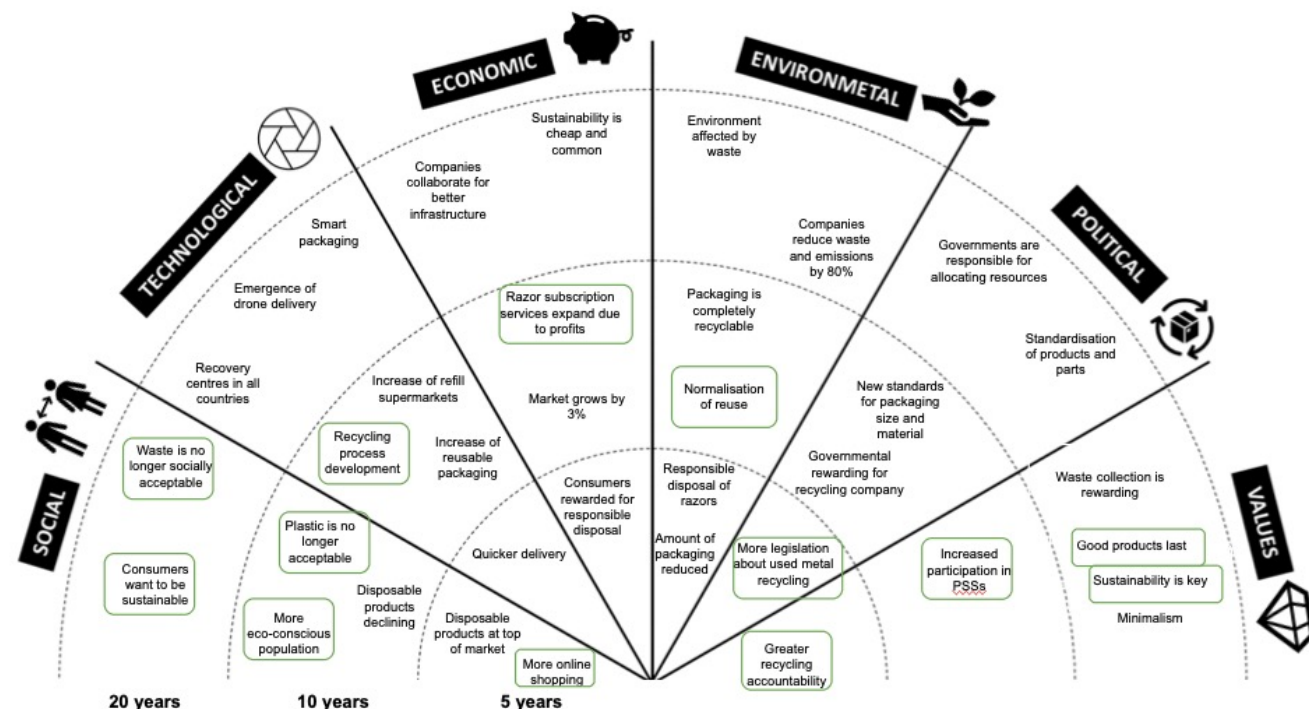
Occupation: Student

Attitude towards sustainability: Linda is very eco-conscious and makes sustainability a core value in her life. She's knowledgeable about climate change and adopts a vegan lifestyle. She is trying to become zero waste.

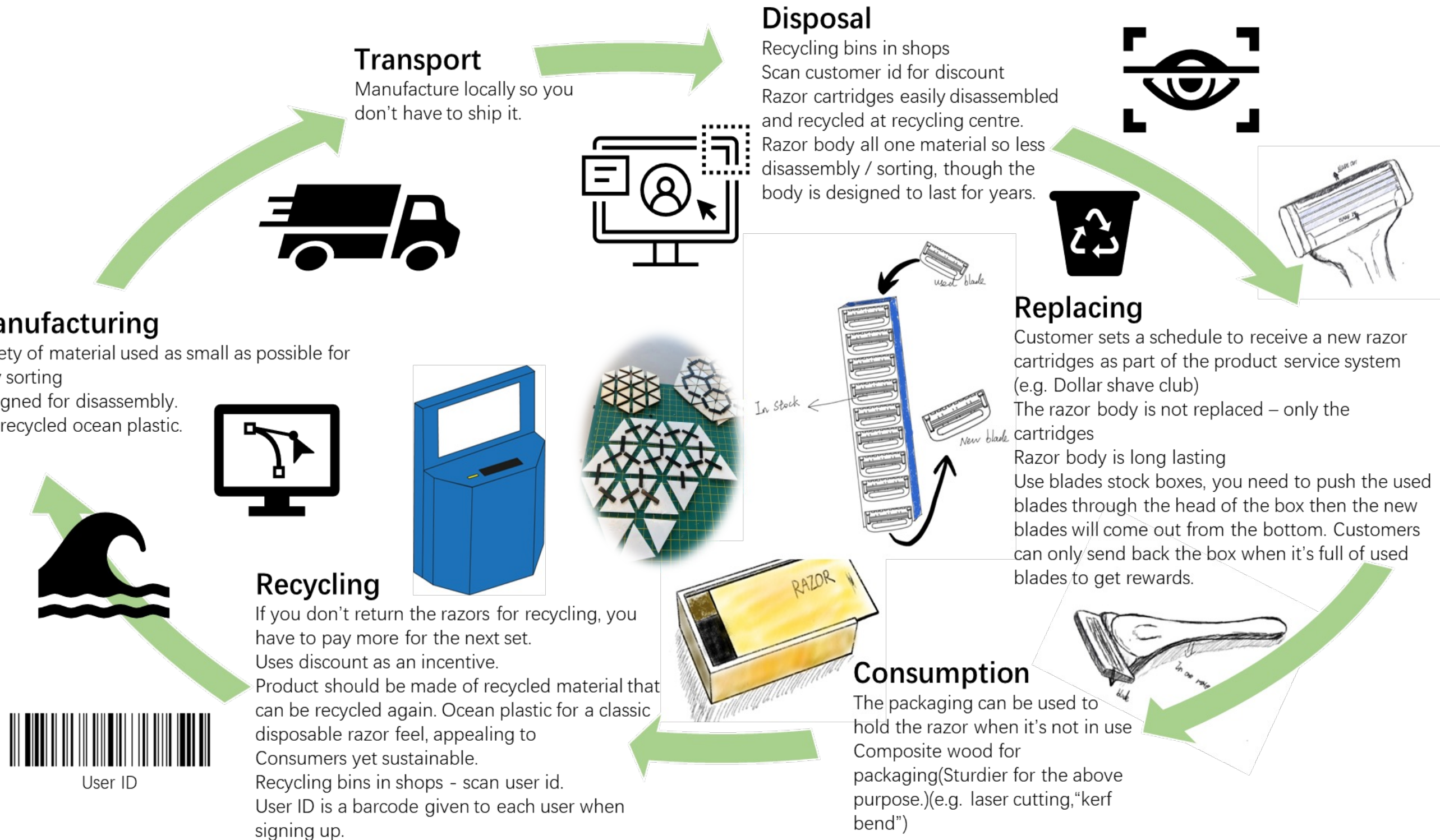
Other factors:

Linda grew up surrounded by technology and is very comfortable navigating through the web, apps and various gadgets.

Why Linda chooses ModuSHAVE: ModuSHAVE works well for Linda as it fits perfectly into her busy life as a student. She goes shopping weekly and when her box is full of razors the drop off boxes are very convenient for her. As a student, she is very wary of her finances, so she really appreciates the discount given for recycling.



System design



Stakeholders

Manufacturer and supplier	Consumer	Direct transport to consumers	Recycler
Supplies us with razors, batches are always relative to demand and periodic inspection is carried out to ensure minimal waste of raw material - Matches supply to demand - Developments leading to efficient reuse of returned materials	Purchases subscription that provides them with monthly supply of razors, with benefits that come with recycling through a membership card that records amount recycled. Incentivised to return used razors for recycling	Monthly direct deliveries to subscribed customers. - Reduces needed storage - Promotes Just in time production, reduces wastes on raw material and over stocking - Reduces carbon emission by only transporting products when needed.	Carbide(The blades) is recycled through a series of grinding and chemical treatments. Recycling plants receive the used cartridges directly from the recycling boxes. - Grounded and chemically treated - The plastics(PET)

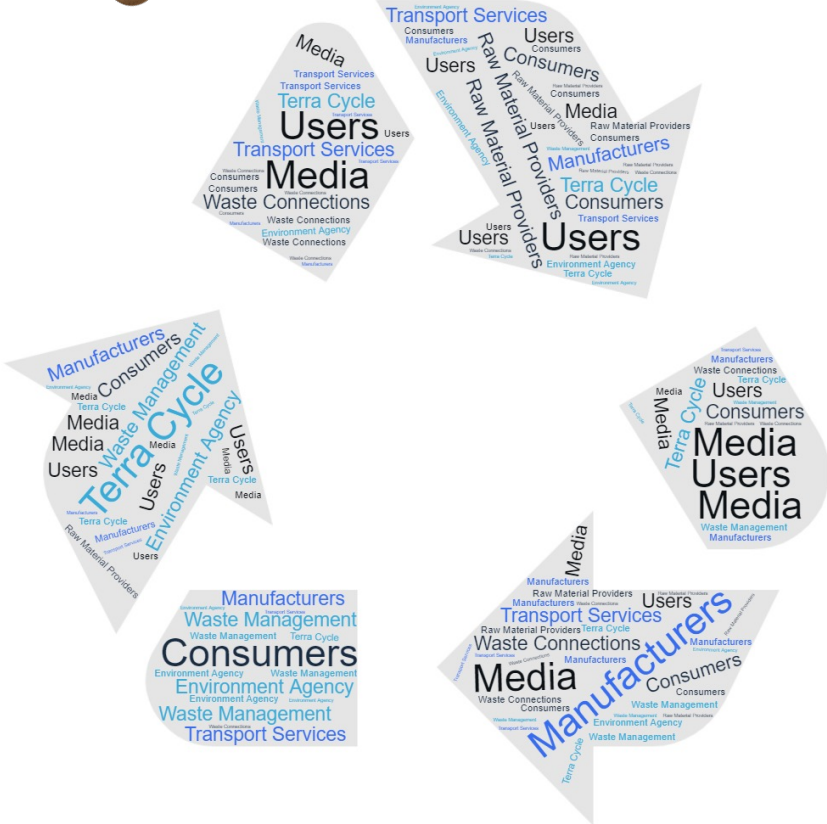
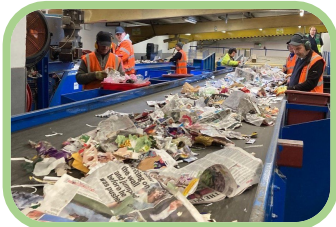
Indirect

Policy Makers

For companies who strive to be more eco-friendly, tax reliefs should be provided. Companies that have less sustainable business models would then have an incentive to become more environmentally aware.

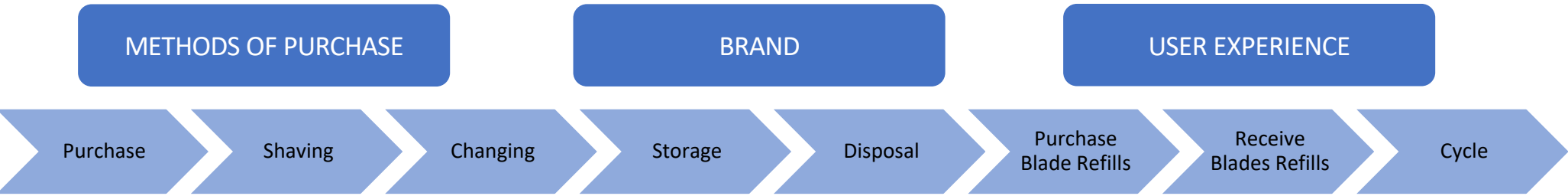
Indifferent

This includes multiple individuals and services, such as the workers for the recycling plants, transport services and manufacturers as well as raw material suppliers. Their efforts are vital to our system and have to be paid in exchange.

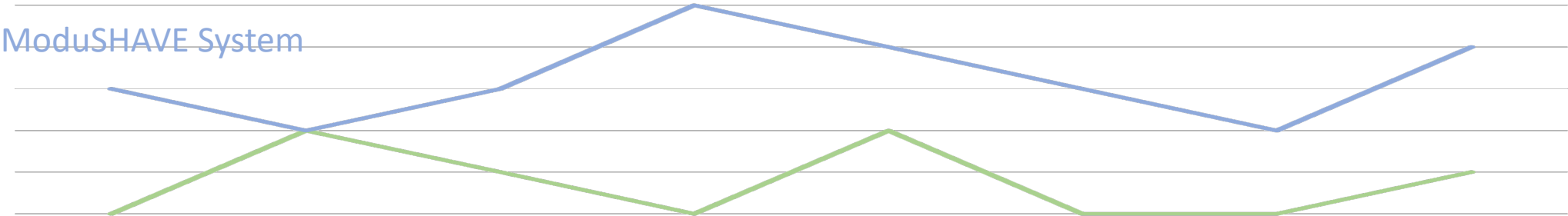


User Analysis

Users of Current Razor System and ModuSHAVE System are different in daily using habits, ModuSHAVE System creates a kind of using method which enables users feel senses of achievement and responsibility.

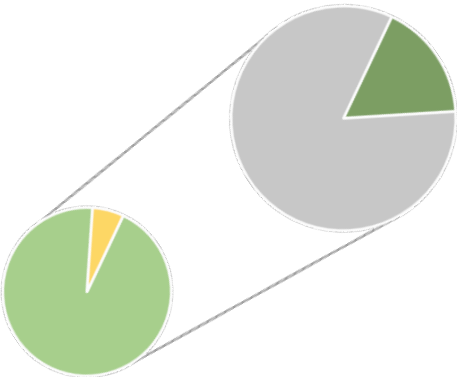


ModuSHAVE System



Current System

You need to buy on-site or online, and you need to find the corresponding product every time	Plastic or metal handle	It needs to be aligned with the buckle, there is a certain risk of cutting the finger	Some will provide a plastic lidless box, which is easy to lost	Disposed of directly in the trash, not directly recycled to the razor company	The same size razor blade needs to be ordered online, which is expensive	Waiting for the courier to arrive, unpack and dispose of new packaging waste	The low sense of participation in sustainable development generates a lot of unnecessary packaging waste
Subscribe at the beginning, and then you only need to pay on a regular basis	Handle made of composite wood	Rotary screw-in design for greater safety and simplicity	There are individual matching wooden boxes to ensure that every used or unused blade is never lost	Recycling point at the nearby supermarket, scan the code to drop in the box, and get the points reward	Regular mailing according to the usage habits set by the user, a discounted price is available if help to recycle	Familiar storage box without excess packaging	You can check the recycling process of used blades through APP, and get a sense of responsibility and achievement
Simplified packaging no longer requires ink printing, and a lot of pollution is reduced, but it will not be able to attract consumers to make purchases in physical stores.	The wooden grip can be textured to increase friction without resorting to rubber materials that are difficult to recycle.		The blade is designed with one-button removal, and the material is the same plastic as the blade holder, which can be directly recycled together.	With only three materials, the entire product can be easily sorted without damaging any structure, greatly simplifying the recycling process.	The packaging can be used repeatedly without discarding under certain conditions, and the subscription system brings users a better after-sales experience.		The mobile app can see the recycling process of used blades, making the concept of sustainability a driving force for visualization.



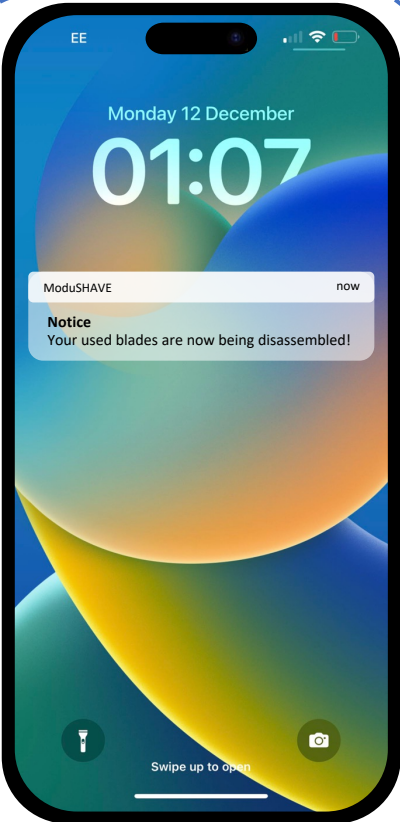
83% of users do not consider the sustainability of the product when choosing a shaver, and many products on the market do not advertise this feature on the packaging. But when the concept of sustainability was mentioned to

respondents, **78%** of users indicated that they would prefer to buy an environmentally friendly shaver and would take this feature into consideration when making future purchases. But at the same time, the respondents also said that the concept of sustainability is not mentioned on the packaging of many razor brands.

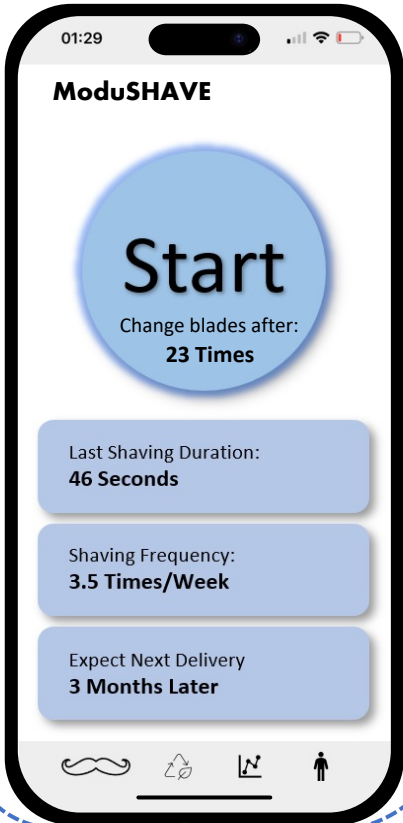


Data processing and the app

To improve the user experience and visualize rewards, ModuSHAVE has developed an app that allows subscription-based users to enjoy the fulfillment and satisfaction of sustainability with the simplest setup.



Whenever the user starts shaving, on the shaving interface, the user can tap the start button and tap again after the shaving is complete. The app will automatically record the shaving time and calculates the time and delivery time of the next blade replacement based on user habits.

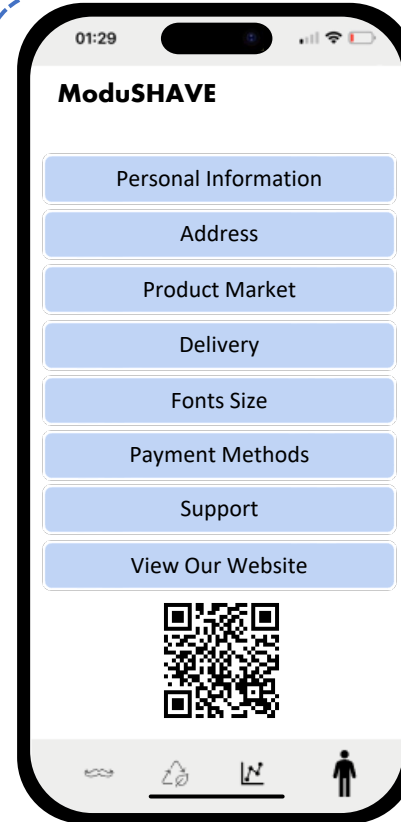
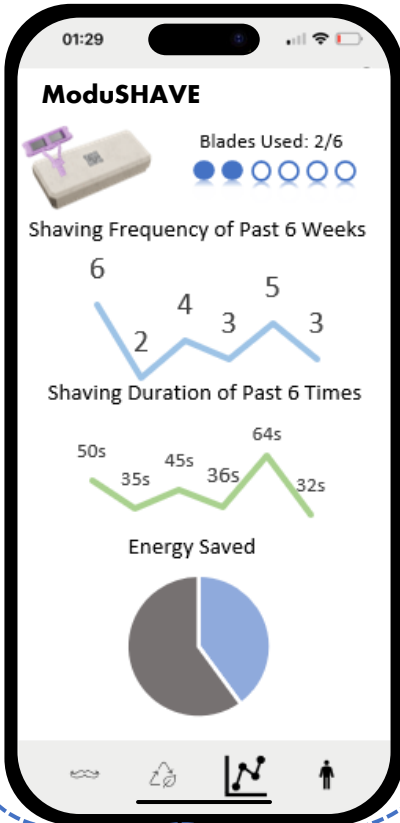


The ModuSHAVE app reminds users to shave at a frequency they set to set. And every time the storage box owned by the user enters the next stage of recycling, a reminder is received. This reminder forms a visual proactive feedback mechanism, allowing users to feel more involved in the sustainability process.



On the sustainability interface, users can see where they are putting their recycled blade cartridges in. After each stage is completed, the user will be rewarded with points, which can be used to purchase products in the product market.

On the statistics interface, you can see the number of replaceable blades, shaving frequency and duration statistics, which is convenient for users to control their usage habits. The gray part of the energy save statistic is the energy consumption generated by using other competitors, and the blue part is the energy consumption generated by using ModuSHAVE.



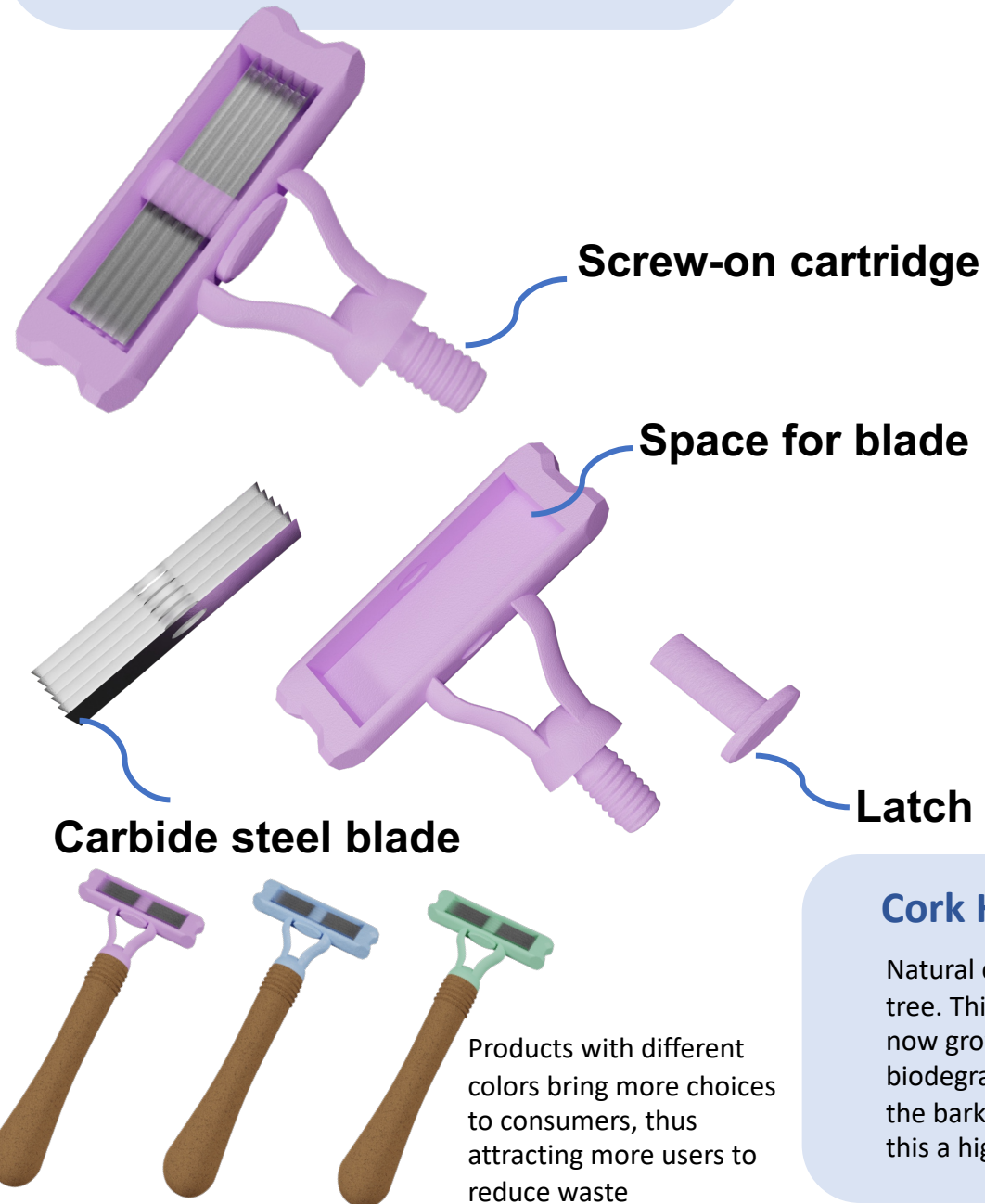
In the personal interface, you can edit personal information and other personalized settings, and in the address function, users can view the nearest recycling points nearby. When users have any suggestions for improvement or feel inconvenient about any link, they can use the support key to contact customer service. If the QR code on the storage box cannot be scanned at the time of recycling, you can also find the QR code here.

Product and packaging design

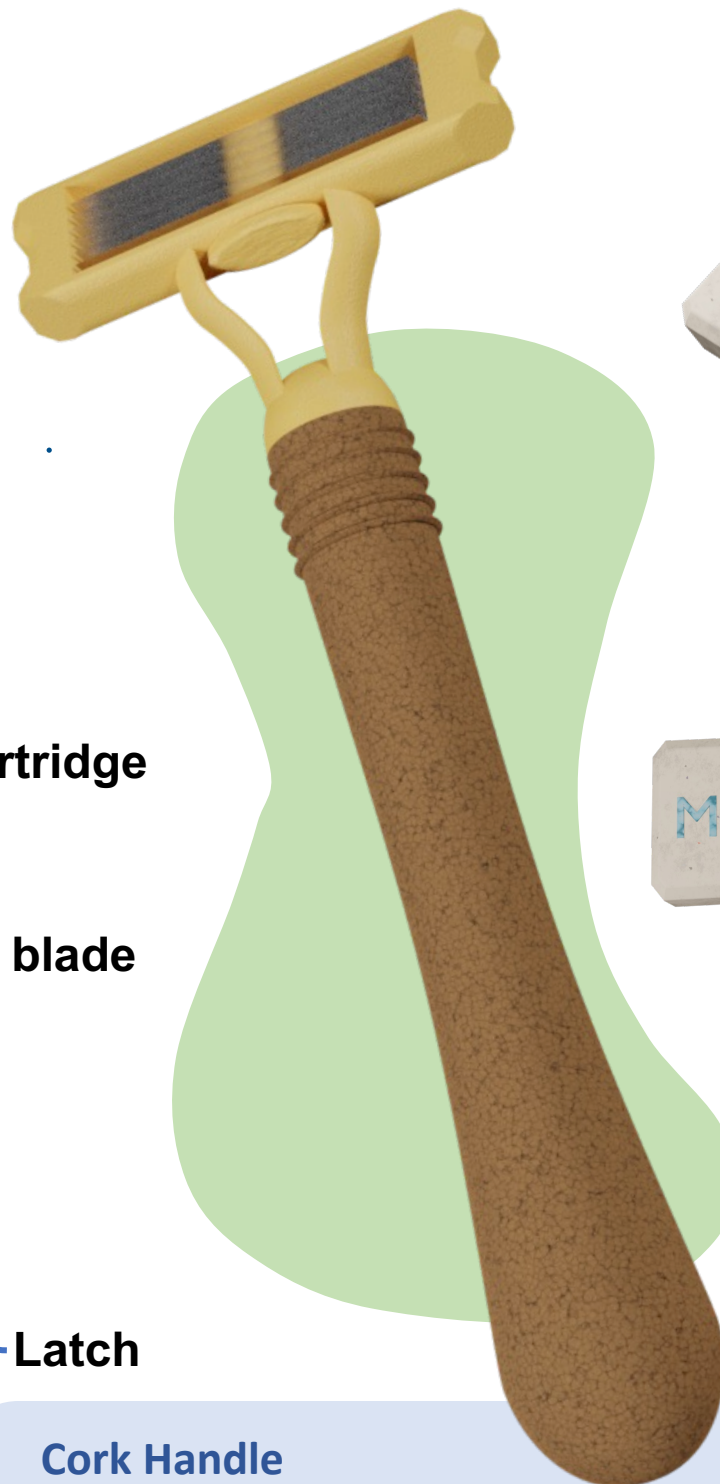
Razor head

The shaver head is mainly composed of three parts, carbide steel blade, the PET cartridge and a latch in the middle. Three-part cartridge designed to be quickly disassembled by professionals. Reducing the materials used in the product would increase the efficiency of recycling processes and pollution.

The picture below shows the details of our design.



Products with different colors bring more choices to consumers, thus attracting more users to reduce waste.



Cork Handle

Natural cork is made from the bark of the cork tree. This is a large tree originally from Asia that is now grown all over the world. It is 100% biodegradable and can be harvested by stripping the bark and without damaging the tree, making this a highly sustainable material.

QR Code

Scan the QR code on the box with our app to receive membership points. You may then view your accumulated points on the app and receive discounts and benefits when you recycle.



Packaging

The wood packaging is 100%. These processed wooden boxes are moisture-proof and anticorrosive, which can be used for both packaging boxes and storage boxes.

The boxes are separated into two parts, one for the razor and another one for the replace-razor head. The razor head box with its QR code would be used to collect the products.

Every 15th day of the month, used components will be collected and transported to the recycling plants, where every part is disassembled, sorted, and disinfected.

When users purchase our products, they can choose the razor or the replace razor-head. The box used for the razor handle will be returned to the factories to collect waste PET and steel and produce new razors using those materials. Reduce the waste of raw materials as much as possible. But if consumers still choose to throw away the product, razors with nearly 100% degradable materials still have no impact on the environment.

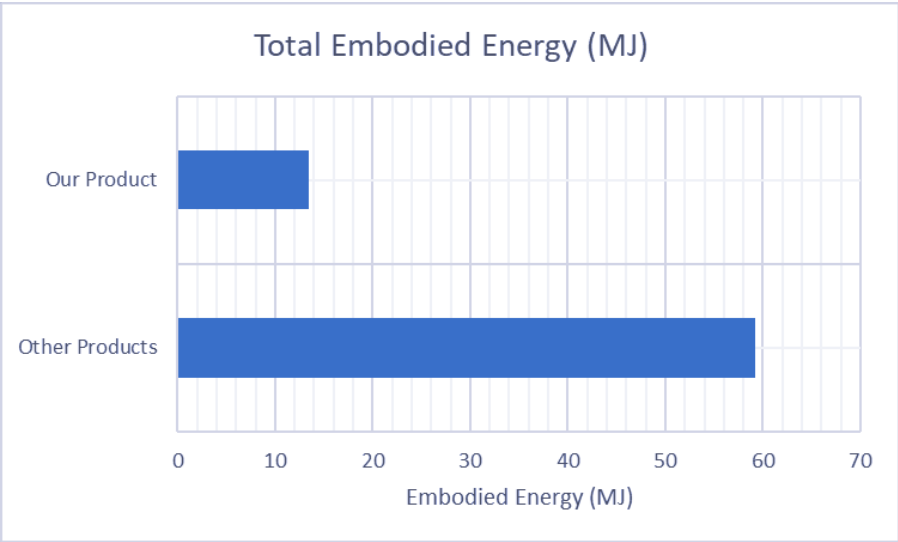
Environmental impact analysis

Our Product

Phase	Energy (MJ)	Energy (%)	CO2 footprint (kg)	CO2 footprint (%)
Material	13.3	98.7	0.106	89.7
Manufacture	0.0307	0.2	0.0023	2.0
Transport	0.0139	0.1	0.001	0.9
Use	0	0.0	0	0.0
Disposal	0.127	0.9	0.00887	7.5
Total (for first life)	13.4	100	0.118	100
End of life potential	-12.9		-0.0791	

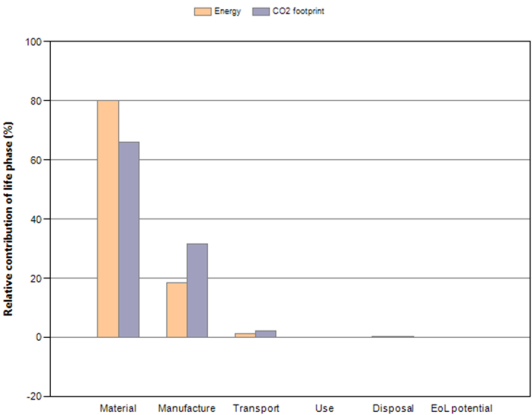
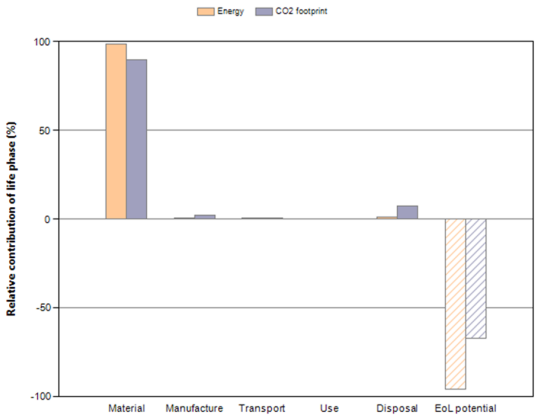
The analysis compares the environmental impact of ModuSHAVE to a generic manual shaver currently on the market.

Overall, we see a massive reduction of over 40MJ per razor, with 13.36 million people using refillable razors in 2020, the savings amount to 53.44 TJ of energy.



Current Products

Phase	Energy (MJ)	Energy (%)	CO2 footprint (kg)	CO2 footprint (%)
Material	47.4	80.1	1.71	66.0
Manufacture	10.9	18.4	0.818	31.6
Transport	0.755	1.3	0.0543	2.1
Use	0	0.0	0	0.0
Disposal	0.137	0.2	0.0096	0.4
Total (for first life)	59.2	100	2.59	100
End of life potential	0		0	



Materials Energy Reduction: 45.8MJ

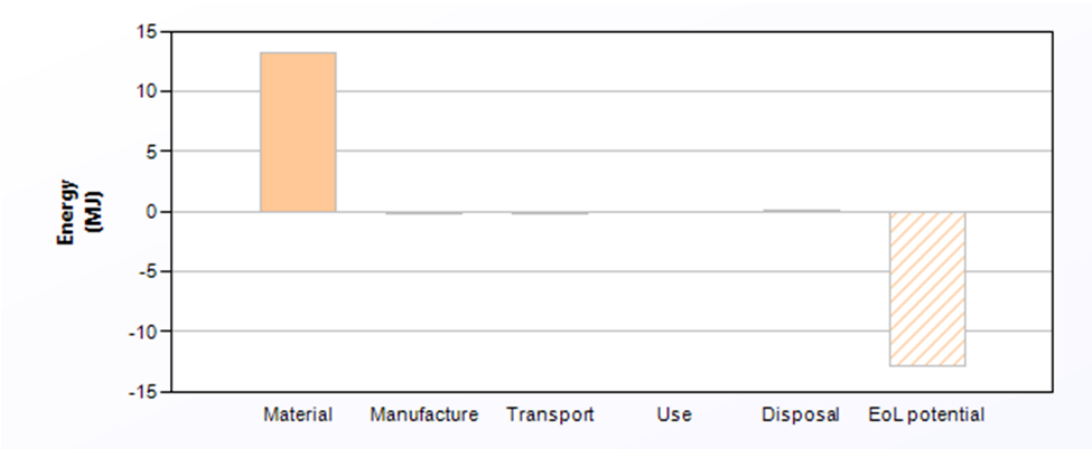
By using the most recycled plastic, PET, and reducing the variety to only one, we were able to reduce the embodied energy massively. Through our system, every component is recycled, down to the blade. This can be attributed to the incredibly easy disassembly of the razor.

Footprint Reduction: 95%

From our manufacturing plants, we produce according to the demand each month, using data collected by our app. The packaged products are shipped off to our depots around the country, where they are temporarily stored and quickly distributed to where they need to go. The reduction in storage, stock and transportation massively decreases the CO2 footprint

End of life: 96% of embodied energy is recovered

All parts are collected and sent directly from recycling points to recycling plants, these are disinfected, and treated according to wear of the component. Products emerge once again from recycled materials. The recovery rate can also be attributed to increased incentives for recycling thanks to our benefits and discount program.



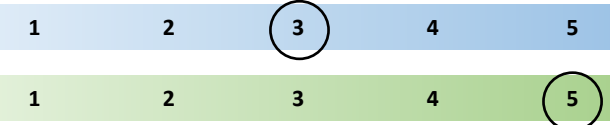
Concept validation and analysis

User validation:

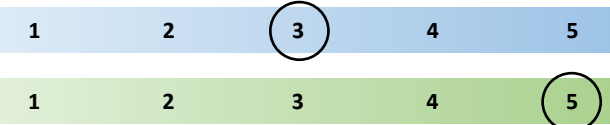
Three users were interviewed about their opinions on the ModuSHAVE system concept when compared to using their current razor and the TerraCycle recycling scheme.



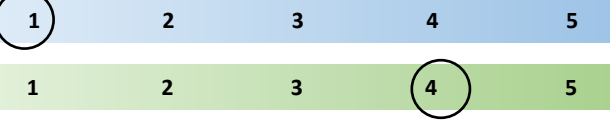
The system is sustainable:



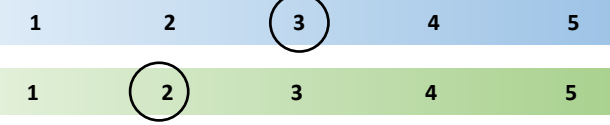
The system is easy to use:



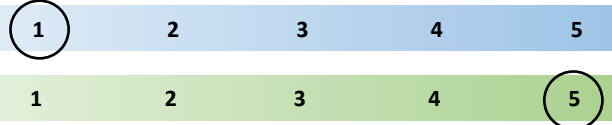
The system is convenient:



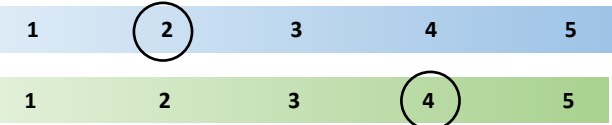
You are likely to overconsume:



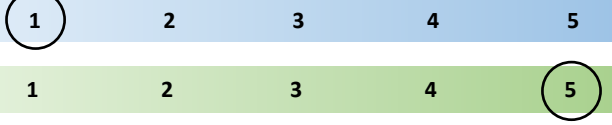
The recycling instructions are clear:



Disposal is easy:



You feel motivated to recycle:



The satisfaction ratio for the current system compared to ModuSHAVE is 3:8. This suggests ModuSHAVE users would be 267% more satisfied than users of the current system.

References:

Fortune Business Insights. *Disposable Razor Blades Market Size, Share & Trends | Forecast [2027]*. [online] [www.fortunebusinessinsights.com](https://www.fortunebusinessinsights.com/disposable-razor-blades-market-102561). Available at: <https://www.fortunebusinessinsights.com/disposable-razor-blades-market-102561>. Accessed December 2022

Lightfoot. (2018). *How Much CO2 Does a Car Emit Per Year?* | Lightfoot. [online] Available at: <https://www.lightfoot.co.uk/news/2017/10/04/how-much-co2-does-a-car-emit-per-year/>. Accessed December 2022

TerraCycle. (n.d.). *The Gillette® Recycling Envelope* | TerraCycle® UK. [online] Available at: <https://www.terracycle.com/en-GB/brigades/gillette-envelope-uk>. Accessed December 2022

AAA Polymer. (2020). *A Simple Guide to Polypropylene Recycling for Businesses*. [online] Available at: <https://www.aaapolymer.com/a-simple-guide-to-polypropylene-recycling-for-businesses/>. Accessed December 2022

Design for X:

Design for function:



The app is very intuitive and easy to navigate. Information is displayed clearly and reminders allow the user to benefit even outside of the app.



The razor is designed to be reminiscent of a classic disposable razor. 75% of consumers prefer to use disposable razors, which means they will be familiar with and enjoy using the product.

Design for assembly:



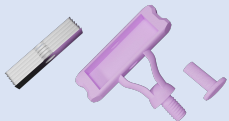
The packaging consists of one material, so only needs one mould. The packaging slots into itself removing the need for fasteners. The razor is comprised of few parts that can be easily assembled.

Design for aesthetics:



ModuSHAVE razors are available in a multitude of colours to suit all user preference. It has a sleek and simple design.

Design for manufacture:



The razor is manufactured from 100% recycled material. The head is made from ocean plastic, the blades are steel and the handle is wooden. If the razor components are undamaged, they will be disinfected and reused, reducing the amount of manufacturing needed. The handles are designed to be long-lasting so users can reuse them for life. This means fewer handles will need to be manufactured. In this scenario manufacturing techniques have advanced, using up to 50% less energy and minimising waste.